

Anna University

REGULAR EXAMINATION ? 2023

Discrete Mathematics ? Regular Examination, 2023

Subject Code: CS305 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Set Theory and Relations)

1. Write short notes on Set Theory and Relations. [12 Marks]
2. Explain Set Theory and Relations with a neat diagram/sketch and a suitable example. [12 Marks]
3. Compare and contrast the key concepts involved in Set Theory and Relations. [12 Marks]

SECTION B (Unit 2: Mathematical Logic)

4. Explain Mathematical Logic with a neat diagram/sketch and a suitable example. [12 Marks]
5. Compare and contrast the key concepts involved in Mathematical Logic. [12 Marks]
6. Solve a problem related to Mathematical Logic. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: Graph Theory)

7. Compare and contrast the key concepts involved in Graph Theory. [12 Marks]
8. How is Graph Theory implemented? Give an algorithm / procedure and explain. [8 Marks]
9. Describe the working principle of Graph Theory with necessary equations. [12 Marks]

SECTION D (Unit 4: Combinatorics)

10. Define Combinatorics. Explain its significance in Discrete Mathematics with a suitable example. [12 Marks]
11. Discuss the engineering applications of Combinatorics. [12 Marks]
12. Write short notes on Combinatorics. [12 Marks]

SECTION E (Unit 5: Algebraic Structures)

13. List the advantages and limitations of Algebraic Structures. Explain any two in detail. [8 Marks]
14. Define Algebraic Structures. Explain its significance in Discrete Mathematics with a suitable example. [8 Marks]
15. Explain Algebraic Structures with a neat diagram/sketch and a suitable example. [12 Marks]

*** End of Question Paper ***